
Interim Report: Graph Neural Network-Based Predictive Modeling for Supply Chain Resilience on a Multi-Modal Network

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Abstract

This progress report presents the development of a Graph Neural Network (GNN)-based framework aimed at enhancing supply chain resilience by addressing limitations in traditional tools that fail to capture disruption propagation and network complexities. The project involves modeling supply chains as multi-modal graphs, simulating disruption scenarios, implementing advanced GNN architectures such as Graph Attention Networks (GAT) and Heterogeneous GNNs (HeteroGNN), and creating actionable insights through an interactive dashboard. Significant progress has been made in setting up the development environment using tools such as PyTorch Geometric, NetworkX, and Weights & Biases. The front-end template for the visualization board is created and ready to be integrated into the back-end infrastructure. Data preprocessing, including cleaning, integration, and feature engineering, is being refined, though delays have been encountered due to challenges such as data accessibility and variability. Preliminary findings from the expected result suggest that GNN models, particularly GAT (Multi-Head), are expected to outperform traditional machine learning methods with an anticipated accuracy improvement of 10–20%. Future work will focus on finalizing graph construction, simulating disruptions, and training GNN models to evaluate and validate the framework's effectiveness. This report highlights the project's progress, challenges, and next steps toward achieving its objectives of improving supply chain vulnerability identification and decision-making.

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List of Abbreviations

1. **AI**: Artificial Intelligence
2. **BCI**: Business Continuity Institute
3. **CSV**: Comma-Separated Values
4. **FERF**: Feature-Enriched Random Forest
5. **GAT**: Graph Attention Network
6. **GCN**: Graph Convolutional Network
7. **GNN**: Graph Neural Network
8. **HeteroGNN**: Heterogeneous GNNs
9. **KPI**: Key Performance Indicator
10. **ML**: Machine Learning
11. **RF**: Random Forest
12. **SCG**: Supply Chain Graph
13. **SVM**: Support Vector Machine
14. **W&B**: Weights & Biases

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1. Introduction

1.1 Background

Supply chain resilience refers to the capacity to anticipate, adapt to, and recover from unexpected disruptions, ensuring continuity and stability [1]. Frequent disruptions can be caused by geopolitical events, climate change, and logistical challenges, resulting in significant economic losses, inefficiencies, and delivery delays [2]. In recent years, modern platforms, for instance Kinaxis Maestro, Oracle Fusion Cloud, and Blue Yonder's Luminate use AI/ Machine Learning (ML) -based forecasting to predict and manage supply chain risks [3], [4], [5]. Similarly, risk management tools for example Interos, Everstream Analytics, and Resilinc focus on mapping supply chain networks, risk analysis and monitoring real-time events [6], [7], [8].

However, while these platforms excel prediction and monitoring, they lack the ability to understand how disruptions spread across interconnected networks; they often fail to capture how disruptions propagate across interconnected networks. For example, if a major port shuts down due to a storm, these tools may not fully account for how the disruption impacts transportation hubs or suppliers [9].

1.2 Project Motivation

Kinaxis Maestro, Oracle Fusion Cloud, and Blue Yonder's Luminate rely on disjointed data structures [3], [4], [5] treating supply chains as isolated data points rather than interconnected networks. This approach overlooks the inherent complexity of supply chains, where disruptions—such as delays in raw material shipments—can ripple downstream through transportation hubs, production lines, and distribution networks, ultimately causing widespread shortages. Capturing these interdependencies is crucial for identifying weaknesses and preventing cascading failures but is lacking in existing platforms.

Graph Neural Networks (GNNs) have emerged as a powerful algorithm for analyzing interconnected systems. In graph theory, a graph represents a network where nodes denote entities such as suppliers, warehouses, or transportation hubs, and edges capture the relationships or dependencies between nodes, such as transportation routes or financial links. GNNs leverage graph structures to model dependencies, capturing both individual attributes and complex interconnections [10].

Studies demonstrate that GNN-based approaches outperform traditional machine learning models, achieving 10–30% higher accuracy in regression and classification tasks and 15–40% better performance in anomaly detection on supply chain benchmarks such as the SCG dataset [11]. Additionally, research by Kayalvizhi *et al.* [1] demonstrated that GNNs improved disruption prediction accuracy by 10 - 20% compared to traditional machine learning methods such as SVM and Random Forest [1]. Beyond improved accuracy, GNNs also enhance interpretability by identifying critical bottlenecks, such as high-risk ports and logistics hubs, enabling proactive mitigation of cascading failures [1].

Despite these advancements, the adoption of GNNs in supply chain resilience analytics remains limited due to three primary challenges: the lack of publicly available datasets labeled with disruptions, difficulties in representing the complex and dynamic nature of supply chains as graphs, and the limited ability to translate GNN predictions into actionable decisions [1], [11]. This project aims to develop a GNN-based framework to enhance supply chain resilience by addressing these challenges and improving vulnerability identification, disruption impact prediction, and proactive decision-making.

1.3 Objectives

The objectives of this project are to enhance supply chain resilience through the development of an attention-based Graph Neural Network (GNN) pipeline. The key objectives are:

- **Model Supply Chains as Multi-Modal Graphs:** Represent supply chains as graphs with nodes (e.g., suppliers, warehouses) and edges (e.g., transportation routes), integrating operational data to reflect real-world complexity [1].
- **Simulate Disruption Scenarios:** Analyze the impact of disruptions by modifying graph attributes and evaluating critical metrics.
- **Train Advanced GNN Architectures:** Implement and evaluate models like Graph Attention Networks (GAT) and Heterogeneous GNNs (HeteroGNN) to predict node-level and edge-level resilience, identifying vulnerabilities.
- **Generate Interpretable Insights:** Provide actionable insights by identifying critical nodes and edges to improve resilience.
- **Develop an Interactive Dashboard:** Create a tool to visualize supply chain graphs, simulate disruptions in real time, and support risk analysis.

1.4 Scope and Key Deliverables

This project focuses on designing, developing, and evaluating a pipeline for predicting supply chain resilience using advanced GNN models. The scope includes modeling supply chains as directed, multi-modal graphs, where "directed" indicates that edges represent a specific direction flow (e.g., from supplier to warehouse). These graphs incorporate attributes including capacities of warehouse, costs, and lead times to reflect real-world complexity. Simulations of disruptions, including capacity reductions, route delays, and port closures, will assess their impact on service levels, costs, and delays. Advanced GNN architectures, including GAT and HeteroGNN, are employed to analyze data and provide actionable insights to supply chain vulnerabilities.

Key deliverables include:

- A detailed graph-based representation of supply chains.
- Labeled datasets generated from disruption simulations.
- Trained GNN models, including variations such as GAT and HeteroGNN.

- An interactive dashboard to visualize supply chain graphs, simulate disruptions in real time, and conduct risk analysis.

1.5 Outline

The remainder of this report is structured as follows: Section 2 presents the project's methodology, detailing the procedures undertaken and providing a rationale for the chosen approach. Section 3 discusses the results achieved to date, highlighting the current progress and key findings. Finally, Section 4 concludes the report, summarizing the project's outcomes and providing insights for future work.

2. Methodology

This section outlines the methodology for designing, implementing, and evaluating an attention-based Graph Neural Network (GNN) pipeline to enhance supply chain resilience. The process is divided into seven subsections: 2.1 Setup and Initialization, 2.2 Data Acquisition and Preparation, 2.3 Graph Construction, 2.4 Simulation of Disruption Scenarios, 2.5 Visualization and Simulation Tool, 2.6 Modeling with Graph Neural Networks, and 2.7 Finalization.

2.1 Setup and Initialization

The development environment for graph-based machine learning was configured using libraries such as PyTorch Geometric and NetworkX to ensure a robust and efficient workflow. PyTorch Geometric was chosen for implementing and training GNN as its availability of prebuilt for advanced architectures such as Graph Attention Networks (GAT) and Heterogeneous GNNs (HeteroGNN), along with its flexibility for experimentation and alignment with cutting-edge practices, especially for the project's graph scale [11], [12], [13]. NetworkX was employed for graph construction and analysis, as it offers extensive capabilities for creating and validating graph structures. Graphs created in NetworkX were subsequently converted into PyTorch Geometric-compatible formats, enabling seamless integration into the GNN training pipeline [1], [11].

Weights & Biases (W&B) was utilized to enhance the development process through advanced visualization tools, real-time monitoring, and seamless collaboration. It supports systematic tracking of experiments, including hyperparameter tuning, training metrics, model versions, and advanced visualizations, ensuring reproducibility and informed decision-making throughout the project lifecycle [11], [13]. Additionally, a version-controlled GitHub repository was established to manage the codebase. This ensured reproducibility, maintained a clear change of history, and provided a centralized backup for the project. Together, these tools form a robust and flexible development environment, aligning with industry best practices for machine learning and reproducible research.

2.2 Data Acquisition and Preparation

Section 2.2.1 describes the acquisition or synthesis and preprocessing of both synthetic and public supply chain datasets, while Section 2.2.2 details the feature engineering steps used to derive graph-structured inputs and node-level resilience labels for GNN-based analysis.

2.2.1 Data Acquisition

Data acquisition in this study follows a two-dataset strategy, designed to mitigate the limited availability of rich, disruption-labeled real-world supply chain data while enabling both controlled experimentation and evaluation under more realistic conditions.

First, a fully synthetic multi-tier supply chain network is constructed to capture essential structural characteristics commonly reported in the supply chain literature. The network comprises nodes representing supply chain entities distributed across four tiers—suppliers, manufacturers, distributors, and retailers—following the natural flow of goods from suppliers → manufacturers → distributors → retailers [1]. Nodes and edges are assigned application-specific attributes such as capacities, costs, lead times, reliability scores, and historical risk levels. This synthetic dataset provides a controlled environment in which the effects of different disruption types and severities on supply chain resilience can be systematically examined.

Second, the study leverages a public dataset that is related to supply chains and graph neural network applications. These datasets are converted into explicit graph representations and subsequently augmented with additional graph attributes required for the proposed modelling framework. This procedure enables the use of real or semi-real data while obtaining the feature set necessary for GNN-based resilience analysis.

Following acquisition and synthetic data generation, cleaning and standardization are performed using Pandas, a widely adopted Python library for structured data manipulation. Pandas are specifically chosen for its flexibility, efficiency, and extensive functionality, which make it ideal for preprocessing large datasets. Pandas is employed to handle missing values, align timestamps, normalize units, and harmonize schemas across heterogeneous data sources. These preprocessing steps reduce noise, enhance internal consistency, and ensure that the data are suitable for subsequent analysis.

2.2.2 Feature Engineering

On the basis of the cleaned datasets, feature engineering is performed to construct node- and edge-level feature vectors and a node-level target variable. For each node, a feature vector is defined that combines numerical attributes (e.g., capacity, reliability, geospatial coordinates, and pre-computed risk scores) with categorical information (e.g., supply chain role) encoded as one-hot vectors.

For each edge, a feature vector is defined that includes the quantity of goods flowing between supply chain entities and other relationship attributes that may affect resilience.

In addition, a binary resilience label is assigned to each node based on a historical resilience score derived from past disruption records and thresholded to classify nodes as resilient or vulnerable. The detailed definitions of these feature vectors and the target variable are provided in Appendix A (A.2, A.3, A.4).

The cleaned and standardized datasets are stored in **CSV format**, selected for its simplicity, human readability, and compatibility with the software stack employed in this project. Finally, a comprehensive data dictionary is developed to document all variables and attributes across both datasets, including node and edge types, feature definitions, and units of measurement, ensuring transparency and traceability for future use.

2.3 Graph Construction

The system architecture will represent supply chains as directed, multi-modal graphs, a robust approach for capturing the complexity and interconnectedness of supply chain operations (see Figure 1, Appendix B) [1], [11]. This representation is particularly well-suited for supply chain analysis, as it allows for simultaneous modeling of diverse entities (e.g., customers, producers, distributors, and companies) and their interdependencies within a unified framework [1], [11]. By doing so, it will enable a comprehensive analysis of supply chain dynamics, including cascading effects of disruptions, which are critical for evaluating resilience [1].

In the baseline supply chain (Figure 1, Left, Appendix B), entities and flows are represented under normal operating conditions, where material, financial, and reverse logistical flows proceed smoothly. In contrast, the disrupted supply chain (Figure 1, Right, Appendix B) demonstrates the impact of failures such as a closed warehouse, resulting in broken flows, rerouted paths, and cascading effects throughout the network. Disruptions are modeled using dashed red lines for rerouted flows and gray lines for broken connections, clearly illustrating the challenges to supply chain resilience [1], [11].

To analyze these scenarios, graphs were constructed with nodes and edges assigned critical attributes such as capacities, costs, and lead times—key parameters for evaluating supply chain performance under normal and disruptive conditions. NetworkX was used for graph construction and analysis due to its flexibility and compatibility with PyTorch, while Pandas handled data preprocessing to prepare datasets for graph generation. The graphs were then converted into PyTorch Geometric-compatible formats for seamless integration with advanced GNN modeling frameworks. PyTorch Geometric was utilized for its robust capabilities in implementing and training GNN architectures, enabling detailed evaluation of supply chain resilience. The resulting graph objects were stored to support efficient analysis and experimentation [1], [11].

2.4 Simulation of Disruptions Scenarios

Real-world disruption scenarios will be designed to assess supply chain resilience. These scenarios will include, but are not limited to, a 30% reduction in capacity at selected warehouses, a 50% increase in lead times along transportation routes, and temporary port

closures. The disruption scenarios draw insights from the BCI Supply Chain Resilience Report (2024) [2], which identifies key risks, threats, and concerns anticipated by organizations over the next five years, such as supply chain insolvency, transportation delays, and cyber-attacks [2]. These scenarios reflect prevalent supply chain challenges, such as natural disasters, civil unrest, and economic disruptions. These scenarios are aligned with industry trends and risk profiles to ensure validity and relevance [2]. Resilience metrics, including service levels, delays, and operational costs, may be quantified using predefined thresholds (e.g., acceptable delays and cost increase tolerances) to measure the impact of disruptions [11], [13].

Disruptions can be simulated by programmatically adjusting graph attributes, such as costs, capacities, and lead times, within the supply chain model. Test cases will be developed to validate the system's ability to simulate disruptions effectively and measure their effects. The output consists of a labeled dataset of resilience metrics under baseline and disrupted conditions, adhering to industry standards for quality and reliability [1], [11].

2.5 Visualization and Simulation Tool

An interactive dashboard is proposed to provide a user-friendly interface for visualizing the supply chain graph and simulating disruption scenarios. Key functionalities include graph visualization, disruption testing, and risk analysis to identify bottlenecks and vulnerabilities. The dashboard is designed to support non-technical stakeholders by providing actionable insights [2], [4].

In the **Overview** view, the dashboard presents high-level key performance indicators such as overall resilience score, aggregate supply chain risk, and the number of active alerts. A global map visualizes the supply chain network, with nodes colored by risk level and differentiated by role, allowing users to quickly identify geographically concentrated vulnerabilities and critical links.

The **Suppliers** view provides more detailed operational metrics, including the total number of suppliers, count of high-risk suppliers, average on-time delivery rate, and average quality score, again with trend information compared to the previous month. A tabular "Supplier Risk Assessment" panel lists individual suppliers with their ID, name, location, risk level, on-time delivery, and quality score, enabling targeted follow-up on specific entities.

The **Risk Analysis** view summarizes the distribution of risk across the network (e.g., proportions of low, medium, high, and critical risk) via visual elements such as pie charts, and compares current performance against target benchmarks for metrics like on-time delivery, quality rate, lead time, cost efficiency, and flexibility.

Usability testing and iterative stakeholder feedback ensure the dashboard adheres to industry standards for accessibility and functionality. Features such as last-updated timestamps, refresh controls, and report export capabilities further support operational use. The final tool therefore enables data-driven decision-making and proactive risk mitigation by combining

high-level resilience indicators with drill-down analytics on suppliers, regions, and evolving risk events.

2.6 Modeling with Graph Neural Networks

To analyze the supply chain graph, advanced GNN architectures, including GAT and HeteroGNN, will be implemented. GAT is expected to enhance predictive accuracy in identifying vulnerabilities by dynamically prioritizing critical nodes and edges, such as high-risk transportation routes and bottleneck points [1], [5]. HeteroGNN is set to be used to handle multi-modal data, capturing the complex interdependencies within supply chains and predicting disruption impacts across diverse modalities [1], [11].

Baseline models, such as Logistic Regression and XGBoost, are planned to be trained for comparative evaluation to benchmark the effectiveness of GNNs [1], [14]. These models are expected to provide a comparative framework for evaluating the anticipated effectiveness of the GNN architectures. Validation is intended to be conducted using cross-validation techniques, and the models are to be assessed based on established performance metrics, including accuracy, F1 score, and interpretability [12].

To improve the interpretability of the GNN models, attention maps may be generated to visually highlight the critical nodes and edges identified by the models. These maps are expected to facilitate a better understanding of the predictions and provide actionable insights for decision-makers. The attention maps will be further validated through iterative stakeholder feedback to ensure their relevance and practical applicability [5].

Firstly, key performance indicators (KPIs) have been identified to evaluate the system's effectiveness and utility. Predictive accuracy, measured through the F1 score, is used to assess model reliability by balancing precision and recall—an essential consideration for handling imbalanced datasets where disruptions are less frequent. This metric ensures accurate identification of disruptions while minimizing false negatives. Secondly, disruption impact quantification evaluates the system's ability to analyze and mitigate risks by assessing metrics such as economic losses, lead time delays, and service level reductions. Lastly, dashboard usability focuses on ensuring that insights are accessible and actionable for end-users, particularly non-technical stakeholders. Together, these KPIs address both technical and operational goals, contributing to the development of a robust and practical tool for supply chain resilience analysis [1].

2.7 Finalization

The project will conclude with a comprehensive final report summarizing the methodology, findings, and recommendations for enhancing supply chain resilience. Results will be presented to supervisors and examiners. The fully documented and version-controlled codebase will ensure reproducibility and accessibility for future research.

3. Results and Discussion

This section outlines the results of the projects. This section is divided into 5 subsections: 3.1 Work accomplished to date, 3.2 Expected Results, 3.3 Difficulties encountered with impact and possible resolutions, 3.4 Work remaining, 3.5 Plans for the next reporting period with a timeline

3.1 Work accomplished to date

The computational environment for this study has been fully established and tested, incorporating PyTorch Geometric for graph neural network (GNN) modelling, Pandas for data preprocessing, and Weights & Biases (W&B) for experiment tracking. These tools have been configured to support reproducible experimentation and systematic evaluation of the proposed methodology.

In accordance with the two-dataset strategy outlined in Section 2.2, a synthetic multi-tier supply chain network with fixed tiers and node counts has been constructed. Initial node- and edge-level attributes have been assigned, and preliminary disruption scenarios have been specified and encoded to facilitate subsequent resilience analysis. In parallel, relevant public datasets have been identified, acquired, and transformed into preliminary graph representations by defining node and edge types and mapping tabular fields to graph entities.

Both the synthetic and benchmark-based datasets have been cleaned and standardized using Pandas, including the treatment of missing values, harmonization of formats and units, and derivation of key attributes such as warehouse capacities, transportation costs, lead times, and basic disruption or risk indicators. These processed datasets are currently undergoing validation to ensure internal consistency, plausibility of attribute values, and suitability for downstream modelling tasks.

Furthermore, an initial interactive dashboard template has been designed and implemented to serve as the primary user interface for this work. The template provides the structural layout for supply chain network visualization, configuration of disruption scenarios, and the presentation of resilience metrics and risk scores, and will be iteratively refined as modelling results become available.

3.2 Expected Results

The expected outcomes of this project should align with the findings of Kayalvizhi et al. (2025), which demonstrated the effectiveness of advanced Graph Neural Network (GNN) models, particularly Graph Attention Networks (GATs), in supply chain resilience modeling [1]. Based on the performance comparison in the research (see Table 1, Appendix), GAT (Multi-Head) is anticipated to deliver the highest performance, with an accuracy of 93.33%, precision of 92.86%, recall of 100%, and an F1 score of 96.30%. This superior performance highlights its ability to accurately predict vulnerabilities and identify critical nodes and edges in supply chain networks.

Other GNN models, such as GraphSAGE (accuracy: 90.67%, F1 score: 93.02%) and GCN (accuracy: 88.00%, F1 score: 91.07%), are also expected to outperform traditional machine learning models like Random Forest (accuracy: 84.00%, F1 score: 82.76%) and SVM (accuracy: 80.00%, F1 score: 83.87%). These results underscore the advantages of GNNs in capturing complex relationships within supply chain graphs, offering improved predictive accuracy and interpretability.

Overall, the use of GNN architectures, particularly GAT, is expected to enhance supply chain resilience by accurately modeling disruptions, providing actionable insights, and supporting robust decision-making.

3.3 Difficulties encountered with impact and possible resolutions

The project has faced several challenges during the data preprocessing phase, which have affected the timeline and required methodological adjustments. A primary issue has been data accessibility: obtaining external datasets with sufficient granularity and clear links to supply chain disruption events has proven difficult. This has delayed the integration of diverse data sources and slowed preprocessing.

Due to challenges in acquiring sufficiently rich, disruption-labelled real-world supply chain data, we adopted a hybrid data strategy and created two types of datasets. First, we generated a fully synthetic multi-tier supply chain network designed to capture key structural characteristics of real networks (e.g., multi-echelon relationships, varying supplier criticality, and differentiated roles). Second, we incorporated public benchmark datasets from real supply chains that have been previously used for GNN research, and we plan to augment them with additional graph attributes required for our modelling, such as lead times, cost parameters, and risk levels. This strategy enables us to leverage real or semi-real data while still obtaining the rich feature set needed for developing and evaluating our GNN-based resilience framework.

Another significant difficulty is public dataset variability, including inconsistent data schemas, missing values, and undefined or sparsely populated attributes. These issues increased the effort required for feature engineering and standardization, and consequently delayed the construction of the final graph representations. To address this, we implemented automated preprocessing pipelines for data cleaning, imputation, and normalization, ensuring that all attributes are transformed into a format compatible with graph-based modelling.

By introducing this hybrid data strategy and automated preprocessing workflows, the project aims to mitigate further delays and ensure that downstream tasks can proceed in alignment with the revised project timeline.

3.4 Work remaining

The remaining tasks are critical to achieving the project objectives and are organized into several interdependent phases. First, the graph construction and simulation phase will

formalize the supply chain as a multi-modal graph, with nodes representing entities such as suppliers, manufacturers, distributors, warehouses, and retailers, and edges capturing transportation routes and other transactional relationships. Within this phase, disruption scenarios will be systematically simulated by perturbing relevant graph attributes (e.g., capacities, costs, lead times, and availability), thereby generating labeled datasets that characterize how disruptions propagate through the network and affect node-level resilience.

Next, the GNN modelling and analysis phase will implement and train Graph Attention Network (GAT) models with multi-head attention to predict node-level resilience scores. Model performance will be rigorously evaluated against baseline methods—such as Logistic Regression, Random Forests, and other conventional machine learning approaches—using appropriate classification and calibration metrics. Comparative results will be used to assess the added value of the proposed GNN-based methodology for supply chain resilience prediction.

Afterward, the GNN modeling phase will then implement and train the Graph Attention Network (GAT) model using multi-head attention mechanisms to predict node-level resilience. The performance of the model will be rigorously evaluated against baseline approaches such as Logistic Regression and Random Forests to validate its effectiveness.

Subsequently, the dashboard integration and refinement phase will focus on connecting the front-end dashboard with the back-end GNN models and simulation pipeline. This phase will enable interactive visualization of the supply chain graph, configuration and execution of disruption scenarios, and real-time presentation of predicted resilience metrics and risk indicators. Emphasis will be placed on usability and interpretability, ensuring that the tool provides actionable insights for both technical and non-technical stakeholders.

Finally, the final report and documentation will consolidate the entire project by detailing the methodology, results, and practical implications. It will also include a fully documented and reproducible codebase, providing a valuable resource for further research and real-world applications.

3.5 Plans for the next reporting period with a timeline

In the original project timeline (Table 2, Appendix), phases and milestones were designed to follow a linear progression from environment setup and data collection through to modeling and final reporting, with explicit deadlines and estimated hours assigned to each task. All required datasets were initially expected to be collected and validated by October 27, 2025 (40 hours), and data preparation—producing cleaned and preprocessed staging datasets for graph construction—was scheduled for completion by November 17, 2025 (50 hours). As shown in the revised timeline (Table 3, Appendix B), these stages required substantially more time than anticipated, due to the need to source or synthesize additional data, reconcile inconsistent dataset formats, address missing values, and accommodate expanded feature engineering requirements. Consequently, dataset collection and validation were postponed to

November 23, 2025, and the completion of data preparation was deferred to December 23, 2025, with the estimated effort increased from 50 to 80 hours.

Delays in the early phases produced a cascading adjustment of subsequent tasks and a partial restructuring of project phases. In the original plan, graph construction (Phase 3) was scheduled to be completed by December 1, 2025 (40 hours), followed by graph refinement and disruption scenario design between December 15, 2025 and January 12, 2026. In the revised timeline, an intermediate Dashboard Development phase is introduced, with the front-end interface for network visualization, simulation, and risk analysis now targeted for completion by January 19, 2026 (40 hours). Graph construction (Phase 4) is rescheduled to January 30, 2026 (40 hours), while the finalized graph representation (Version 2) and the design of disruption scenarios with labeled resilience metrics are now planned for February 5, 2026 (30 hours) and February 16, 2026 (50 hours), respectively.

Notwithstanding these revisions, the core analytical and reporting milestones remain broadly aligned with the original completion window. Baseline models (e.g., Logistic Regression, XGBoost) and advanced GNN models (e.g., GAT, HeteroGNN) are still scheduled within the February–March 2026 period, with baseline model evaluation now due by February 28, 2026 (50 hours) and advanced GNN modeling retained at March 16, 2026 (80 hours). A new Dashboard Integration phase has been added to connect the functional dashboards to the back-end GNN models by April 12, 2026 (50 hours), while the finalization phase—covering the final report, codebase, and presentation—continues to be due on April 19, 2026 (50 hours). Overall, the revised timeline (Table 3, Appendix) explicitly incorporates the additional effort required for data acquisition and preparation, while preserving the feasibility of the project’s key objectives and final deadline.

4. Conclusion

This progress report outlines the development of a Graph Neural Network (GNN)-based framework aimed at enhancing supply chain resilience by addressing limitations in traditional tools, such as their inability to model disruption propagation and network complexities. The project focuses on modeling supply chains as multi-modal graphs, simulating disruption scenarios, implementing advanced GNN architectures, and creating actionable insights through an interactive dashboard.

Progress to date includes the establishment of a robust development environment leveraging tools like PyTorch Geometric, NetworkX, and Weights & Biases, as well as significant advancements in data preprocessing and standardization and creation of the interface template. Expected results suggest that GNN models, particularly GAT (Multi-Head), have the potential to outperform traditional machine learning approaches, with anticipated accuracy improvements of up to 10–20% over methods like Random Forest and SVM. These results highlight the ability of GNNs to predict vulnerabilities and identify critical supply chain nodes and edges.

However, challenges have been encountered, particularly with data accessibility and variability. These issues have delayed data preprocessing and graph construction. Automated pipelines for data cleaning and imputation have been developed to address these challenges, and efforts to source additional high-quality datasets are ongoing.

Looking ahead, the focus will be on finalizing data preprocessing, completing graph construction, and simulating disruption scenarios. Subsequent phases will include implementing and evaluating advanced GNN models, integrating the model with the interactive dashboard, and validating the system's performance against baseline models. Further research will address limitations related to dataset accessibility, real-time adaptability, and actionable decision-making.

This progress highlights the potential of GNN-based approaches to transform supply chain resilience analytics and lays out the groundwork for achieving the project's objectives in the remaining phases.

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Appendices

Appendix A

This appendix provides formal definitions of the node- and edge-level feature vectors used in the proposed graph neural network (GNN) framework, as well as the construction of the node-level target variable (resilience label).

This appendix provides formal definitions of the node- and edge-level feature vectors used in the proposed graph neural network (GNN) framework, as well as the construction of the node-level target variable (resilience label).

A.1 Graph, Node, and Edge Definitions

We model the supply chain as a directed, attributed graph

$$G = (\mathcal{V}, \mathcal{E}, X, E),$$

where:

- $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ is the set of nodes representing supply chain entities: suppliers, manufacturers, distributors, warehouses, retailers.
- $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of directed edges representing material flows between entities.
- $X \in \mathbb{R}^{N \times d_v}$ is the node feature matrix, where each node $v_i \in \mathcal{V}$ is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^{d_v}$.
- $E \in \mathbb{R}^{|\mathcal{E}| \times d_e}$ is the edge feature matrix, where each edge $(v_i, v_j) \in \mathcal{E}$ is associated with a feature vector $\mathbf{e}_{ij} \in \mathbb{R}^{d_e}$.

In the following sections, we detail the construction of $\mathbf{x}_i$ and $\mathbf{e}_{ij}$, as well as the definition of the node-level target variable.

A.2 Node-Level Feature Vector

For each node $i \in \mathcal{V}$, the feature vector is defined as the concatenation of numerical and categorical attributes:

$$\mathbf{x}_i = \begin{bmatrix} \mathbf{x}_i^{\text{num}} \\ \mathbf{x}_i^{\text{cat}} \end{bmatrix},$$

where $\mathbf{x}_i^{\text{num}}$ denotes the numerical attributes and $\mathbf{x}_i^{\text{cat}}$ denotes the categorical attributes encoded as one-hot vectors

A.2.1 Numerical Node Attributes

The numerical component $\mathbf{x}_i^{\text{num}}$ includes, for example, the following attributes:

1. **Capacity** c_i : Maximum operational capacity of entity i , e.g. measured in units per time period: $c_i \in \mathbb{R}_{\geq 0}$.
2. **Reliability** r_i : A reliability score based on historical performance (e.g., probability of on-time fulfillment or a normalized reliability index): $r_i \in [0, 1]$.
3. **Location coordinates** $(l_x, l_y)_i$: Geospatial coordinates representing the physical location of entity i (e.g., longitude and latitude or projected coordinates): $(l_x, l_y)_i \in \mathbb{R}^2$.
4. **Risk score** rs_i : A pre-computed risk assessment score (e.g., integrating disruption frequency, severity, and exposure factors): $rs_i \in \mathbb{R}_{\geq 0}$.

Collectively, the numerical attributes can be written as:

$$\mathbf{x}_i^{\text{num}} = [c_i, r_i, l_{x,i}, l_{y,i}, rs_i]^\top.$$

All numerical attributes are standardized across the dataset to have zero mean and unit variance:

$$\tilde{x} = \frac{x - \mu_x}{\sigma_x},$$

where μ_x and σ_x denote the empirical mean and standard deviation of attribute x in the training set.

A.2.2 Categorical Node Attributes

The categorical component $\mathbf{x}_i^{\text{cat}}$ encodes discrete attributes as one-hot vectors. The primary categorical attribute is the supply chain role of node i :

$$\text{role}_i \in \{\text{Supplier, Manufacturer, Distributor, Warehouse, Retailer}\}.$$

This is represented via one-hot encoding: $\mathbf{x}_i^{\text{role}} \in \{0, 1\}^5$,

where exactly one element of $\mathbf{x}_i^{\text{role}}$ is equal to 1, corresponding to the role of node i .

Additional categorical attributes (e.g., region, industry segment, ownership type) may be included as needed, each encoded as a separate one-hot vector and concatenated to form:

$$\mathbf{x}_i^{\text{cat}} = \begin{bmatrix} \mathbf{x}_i^{\text{role}} \\ \mathbf{x}_i^{\text{region}} \\ \mathbf{x}_i^{\text{segment}} \\ \vdots \end{bmatrix}.$$

A.3 Edge-Level Feature Vector

For each directed edge $(i, j) \in \mathcal{E}$, representing a relationship (e.g., transportation or supply) from node i to node j , the edge-level feature vector $\mathbf{e}_{ij}$ is defined as:

$$\mathbf{e}_{ij} = \begin{bmatrix} \mathbf{e}_{ij}^{\text{num}} \\ \mathbf{e}_{ij}^{\text{cat}} \end{bmatrix},$$

where $\mathbf{e}_{ij}^{\text{num}}$ and $\mathbf{e}_{ij}^{\text{cat}}$ denote numerical and (optional) categorical attributes, respectively.

A.3.1 Numerical Edge Attributes

The numerical component $\mathbf{e}_{ij}^{\text{num}}$ typically includes:

1. Flow quantity q_{ij} : Quantity of goods or volume flowing from entity i to entity j : $q_{ij} \in \mathbb{R}_{\geq 0}$.
2. Lead time t_{ij} : Transportation or processing lead time between i and j : $t_{ij} \in \mathbb{R}_{\geq 0}$.
3. Transportation cost cost_{ij} : Unit or total cost associated with the link from i to j : $\text{cost}_{ij} \in \mathbb{R}_{\geq 0}$.
4. Edge-level disruption or risk indicator erd_{ij} :
A risk or disruption score associated with the connection (e.g., route reliability or disruption frequency): $\text{erd}_{ij} \in \mathbb{R}_{\geq 0}$.

Hence, $\mathbf{e}_{ij}^{\text{num}} = [q_{ij}, t_{ij}, \text{cost}_{ij}, \text{erd}_{ij}]^\top$,

with each attribute standardized (zero mean, unit variance) over the training set.

A.3.2 Categorical Edge Attributes (Optional)

Where applicable, edges may also include categorical attributes, such as:

- Transport mode (e.g., road, rail, air, sea),
- Route category (e.g., domestic vs. international).

Each is encoded as a one-hot vector and concatenated to form $\mathbf{e}_{ij}^{\text{cat}}$

A.4 Target Variable: Node-Level Resilience Label

The prediction target is a binary node-level resilience label indicating whether each node is resilient or vulnerable under disruption scenarios.

A.4.1 Historical Resilience Score

For each node i , a historical resilience score R_i is computed from past disruption records. This score may aggregate, for example:

- Frequency of disruptions affecting node i ,
- Average recovery time after disruptions,
- Fraction of demand met during disruption periods,
- Severity-weighted impact on upstream and downstream nodes.

Formally, this can be represented as:

$$R_i = f(\text{disruption history of node } i),$$

where $f(\cdot)$ denotes an aggregation function (e.g., a weighted sum or normalized index). The exact formulation of f depends on the available historical data and project-specific design choices.

A.4.2 Binary Label Definition

A binary resilience label $y_i \in \{0, 1\}$ is defined by thresholding the historical resilience score R_i at a chosen threshold τ :

$$y_i = \begin{cases} 1, & \text{if } R_i \geq \tau \quad (\text{resilient}), \\ 0, & \text{if } R_i < \tau \quad (\text{vulnerable}). \end{cases}$$

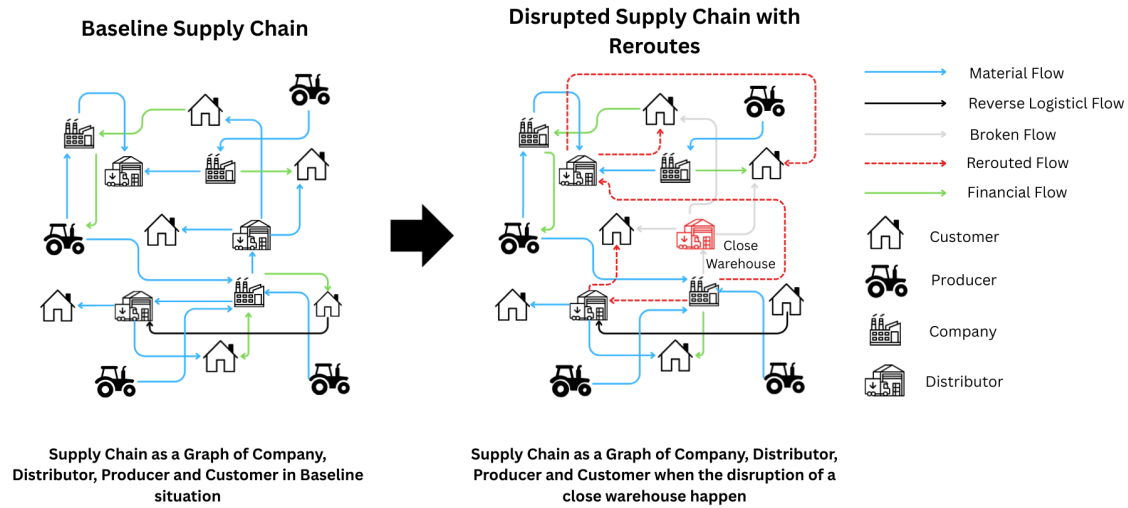
The threshold τ may be selected based on:

- Domain knowledge (e.g., minimum acceptable resilience level),
- Statistical criteria (e.g., median or quantile of R_i),
- Class balance considerations (e.g., ensuring a reasonable ratio of resilient/vulnerable nodes).

The resulting binary labels are used as the supervised learning target for node-level classification in the GNN and baseline models.

Appendix B

Figure 1: Baseline (Left) and Disrupted (Right) Supply Chain Representations



Note: The figure is Self-created, inspired by concepts in [1], [11]. Baseline flows represent normal operations, while the disrupted version highlights broken flows, rerouted paths, and cascading effects caused by a closed warehouse.

Table 1: Performance Comparison of Supply Chain Resilience Prediction Models

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.7867	0.8462	0.7333	0.7857
Random Forest	0.8400	0.8571	0.8000	0.8276
SVM	0.8000	0.8125	0.8667	0.8387
FERF	0.8533	0.8710	0.9000	0.8852
Node2Vec + RF	0.8667	0.8800	0.9167	0.8980
GCN	0.8800	0.8889	0.9333	0.9107
GraphSAGE	0.9067	0.9091	0.9524	0.9302
GAT (Single-Head)	0.9200	0.9130	0.9545	0.9333
GAT (Multi-Head)	0.9333	0.9286	1.0000	0.9630

***Note:** Table 1 presents the performance comparison of supply chain resilience prediction models. Results are sourced from [1]*

Table 2: Original Project Timeline with Phases, Milestones, Deadlines, and Estimated Hours

Phase	Milestone Description	Deadline	Estimated Hours
Phase 1: Setup and Initialization	Environment, repository, and tools fully configured; project objectives, KPIs, scope and risk register finalized.	Oct 13, 2025	30 hours
	All required datasets are collected and validated for further processing.	Oct 27, 2025	40 hours
Phase 2: Data Preparation	Cleaned and preprocessed staging datasets ready for graph construction	Nov 17, 2025	50 hours
Phase 3: Graph Construction	Initial graph (Version 1) constructed with all nodes, edges, and attributes.	Dec 1, 2025	40 hours
Phase 4: Disruption Scenarios	Finalized graph representation (Version 2) after review and refinement.	Dec 15, 2025	30 hours
	Disruption scenarios designed and labeled datasets with resilience metrics generated.	Jan 12, 2026	50 hours
Phase 5: Dashboard Development & Modeling Prep	Functional dashboard built for network visualization, simulation, and risk analysis.	Feb 2, 2026	60 hours
Phase 6: Modeling and Analysis	Baseline models (e.g., Logistic Regression, XGBoost) are trained and evaluated for initial performance metrics.	Feb 16, 2026	50 hours
	Advanced GNN models (e.g., GAT, HeteroGNN) are trained, validated, and insights generated.	Mar 16, 2026	80 hours
Phase 7: Finalization	Final report, codebase, and presentation completed and delivered.	Apr 19, 2026	50 hours

Note: Detailed original timeline of project phases, milestones, deadlines, and estimated hours for the development and evaluation of the GNN pipeline to enhance supply chain resilience.

Table 3: Revised Project Timeline with Phases, Milestones, Deadlines, and Estimated Hours

Phase	Milestone Description	Deadline	Estimated Hours
Phase 1: Setup and Initialization	Environment, repository, and tools fully configured; project objectives, KPIs, scope and risk register finalized.	Oct 13, 2025	30 hours
	All required datasets are collected or synthesized and validated for further processing.	Nov 23, 2025	40 hours
Phase 2: Data Preparation	Cleaned and preprocessed staging datasets ready for graph construction	Dec 23, 2025	80 hours
Phase 3: Dashboard Development	Functional dashboard front-end interface built for network visualization, simulation, and risk analysis.	Jan 19, 2026	40 hours
Phase 4: Graph Construction	Initial graph (Version 1) constructed with all nodes, edges, and attributes.	Jan 30, 2026	40 hours
Phase 4: Disruption Scenarios	Finalized graph representation (Version 2) after review and refinement.	Feb 5, 2026	30 hours
	Disruption scenarios designed and labeled datasets with resilience metrics generated.	Feb 16, 2026	50 hours
Phase 6: Modeling and Analysis	Baseline models (e.g., Logistic Regression, XGBoost) are trained and evaluated for initial performance metrics.	Feb 28, 2026	50 hours
	Advanced GNN models (e.g., GAT, HeteroGNN) are trained, validated, and insights generated.	Mar 16, 2026	80 hours
Phase 6: Dashboard Integration	Functional dashboards are integrated into the back end GNN model.	Apr 12, 2026	50 hours
Phase 7: Finalization	Final report, codebase, and presentation completed and delivered.	Apr 19, 2026	50 hours

Note: Revised timeline of project phases, milestones, deadlines, and estimated hours, reflecting delays in the development and evaluation of the GNN pipeline to enhance supply chain resilience.